release positions are provided, which throttled the flow of air to or from the brake cylinder. The detailed construction of these valves is not particularly important, but they are generally based on the rotary valve principle though some poppet type brake valves can also be found in service.

As a rule the brake cylinders were of standard design, though a small size.

This equipment, is, of course, suitable for single car operation only. For the operation of motor trailer units, with trail cars specially designed for the manufacturer's particular motor car a straight air equipment was provided, having an emergency feature to set the brakes in case of a break-in-two.

With this arrangement the emergency valve does not move with the service application. The only difference between service and emergency brake effect is in the time required for a full brake application, the emergency being much the faster.

It is customary to supply a conductor's valve of standard design in each car, which permits the brakeman or conductor to make an emergency application of the brakes.

The trailer car brake equipment is similar to the motor car brake except that the brake valve and main reservoir pipe are omitted, and the main reservoirs are replaced by one supply reservoir. This tank is charged from the emergency pipe through the emergency valve in release position and furnishes an independent local supply of air for the trailer brake cylinder during an emergency application.

It is, therefore, obvious that with a break-in-two application, both emergency valves assume the emergency position and on the motor car admit main reservoir air to the brake cylinder, which is cut off from the straight air pipe, and on the trailer car, admit supply reservoir air to the brake cylinder, which is also cut off from the straight air pipe.

This equipment permits a very flexible control of the service brakes on a motor trailer unit and provides for automatic application in case of break-in-two, without the triple valve complication and maintenance.

Motor cars in service and building, which are designed to handle a standard passenger coach as a trailer, require a combined automatic and straight air brake equipment. Some of these motor cars have sufficient power to handle a number of standard passenger or freight cars on low speed schedule, or in switching service and, therefore, should be equipped with an equalizing piston type of brake valve.

This brake equipment is somewhat similar to the equipment previously described, except that the emergency valve is replaced with a triple valve, which provides for quick recharge, quick service, graduated release and high emergency cylinder pressure. This triple valve requires the application of one auxiliary and one supplementary reservoir per car, and the following items are also used: Feed valve to automatically maintain a predetermined pressure in the brake system, a double check valve which separates the automatic and straight air systems, so that the operation of either may be obtained without interference. It is, of course, necessary to use additional strainers, cut-out cocks, dirt collectors, couplings, etc.

The main reservoir supply admits air through a cut-out cock to the feed valve which in turn supplies the brake valve. The brake pipe is made up to the brake valve and hose connections, through the necessary strainers and cocks to the triple valve and to the conductor's valve. The straight air application and release pipe connects the brake valve to the double check valve, which in turn is connected to auxiliary reservoir, brake cylinder and triple valve. In addition to the brake pipe and the double check valve connection to the triple valve, are the supplementary and auxiliary reservoir pipes. Air for the compressor governor, signal system and for accessory pneumatic devices is taken directly from the main reservoir. The brake valve for this equipment should be of the rotary valve design, and the straight air service and automatic functions should be detailed in the one valve. The valve handle positions beginning at the left of the quadrant should be release, straight air lap, straight air application, automatic lap, automatic service, handle off, and emergency.

Inasmuch as the straight air system will apply the brakes on the motor car only, it should be used only when the vehicle is in single car service. A cut-out cock is provided in the straight air pipe under the brake valve, so that the straight air system may be cut out when operating a train of two or more cars. During single car operation air is admitted at feed valve pressure through the straight air application and release pipe to the double check valve and from there to the brake cylinder to apply the brakes, which may be graduated on or off in any number of stages, under the direct control of the engineer. This is the most reliable flexible brake control, and for that reason is supplied for single car operation.

When operating a train the straight air feature should be cut out and brake pipe reductions and recharges made in the usual manner. It should be noted, however, that it is impossible to overcharge during normal operation, for the feed valve supplies all air to the brake valve, and there is, therefore, no position where main reservoir pressure can be introduced into the brake pipe. The handle off position is required on double end equipments to prevent manipulation of the non-operative brake valve by unauthorized persons, for it may be located in a position partially exposed to passengers. The emergency position of the brake valve performs the functions with which you are all familiar; namely, to create a rapid reduction in brake pipe pressure, and in addition also to charge the straight air by-pass pipe. If for any reason the triple valve or valves fail to apply, a straight air application will be secured on the motor car, for the by-pass pipe is connected to the straight air pipe at a point below the straight air cut-out cock. This is also a non-return check valve with an orifice, through the valve itself in the straight air by-pass pipe, which reduces the rate of straight air application in emergency, so that the triple valve can still vent brake pipe to brake cylinder, and also freely permits straight air release.

The double check valve has several important functions; first, to permit either a straight air or an automatic application without interference from the non-operative brake system; i. e., to deliver air to the brake cylinder from either one of two sources and seal off the unused source against blow down and leakage; second, to permit the complete release of brake cylinder air when the brake valve is placed in release position and the brake pipe recharged; third, to prevent stuck brakes in case a partial straight air application is made followed by an automatic application. Inasmuch as this double check valve is a new device to most of our members, a detailed description here might not be out of place. It consists primarily of a floating piston valve, one end of which is connected to the normal triple valve brake cylinder pipe tap, the center to the brake cylinder, and the other side of which is connected to the center of a relay piston, so that when a straight air application is made the relay piston unseats an auxiliary reservoir valve, allowing this pressure as well as the straight air pressure to flow directly to the straight air side of the floating piston. With either a straight air or an automatic application, therefore, the floating piston seals off the non-operative systems and passes the